

Artificial Intelligence (AI2002)

Sessional 2 Exam

Date: 09/04/2025

Total Time: 1 hr

Course Instructors

Total Marks: 18

Dr. Fahad, Bilal, Anaum, Nauraiz, Alina, Saif-Ur-Rehman, Yusra, Fizza, Ramsha, Saeeda, Farooq

Total Questions: 3

Roll No

Section

Student Signature

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Attempt All questions.

CLO2: To identify and relate methods of search and practically apply the corresponding techniques.

Q1. Genetic Algorithm

[Marks 6, 4+2, 15 min]

You are given the target string: "HELLO". You are given the initial population below: -

RGYXP,

HQLLA,

XELLO,

HZKLO

A. Use a Genetic Algorithm to evolve the population of strings to match the target. **Describe** the steps of the GA process, including:

1. Initial population generation (given).
2. Fitness function (Evaluate as the number of characters matching in the correct position)
3. Selection (tournament selection select any two pairs and making sure good solutions go to next generation)
4. Crossover (Crossover point at 2)
5. Mutation (Fixed: The gene at index 3 in a selected chromosome is randomly replaced with any letter from a-z)
6. Termination condition

Note: Next population consists of 2 parents from the initial population and 2 offspring.

B. Show how the GA evolves toward the solution after the **first generations** and also **calculate the fitness value** for the evolved generation. Explain

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Q2. Adversarial Search

[Marks 4, 15 min]

A. A game is played between max and min. Draw a tree considering Max as a first player and the branching factor is 2. Given terminal values below, show backed-up values, the best decision available at the root, and branches that will get pruned (explain why they are getting pruned if any).

4, 5, 2, 3, 7, 6, 11, 2, -14, -7, -21, -24, 9, 10, 11, 12.

CLO2: To identify and relate search methods and practically apply the corresponding techniques.

Q3. Constraint Satisfaction Problems

[Marks 8, 4+4, 30 min]

Consider a Constraint Satisfaction Problem defined by $\alpha, \beta, \theta \in \{a, b, c\}$ and constraints $\beta = \theta$ and $\alpha > \beta$ (lexical: 'c' > 'b' > 'a').

A. Demonstrate a step-by-step dry run of the Arc Consistency algorithm. Start by showing the initial queue of arcs, and for each step of the algorithm, clearly indicate the following.

- The arc is being processed.
- Any values removed from the domains of the variables after enforcing arc consistency.
- The updated domains after processing the arc.
- Any new arcs added to the queue as a result of domain reduction.

What are the final domains of variables α , β , and θ after the Arc Consistency Algorithm has completed its execution?

B. You are scheduling 5 classes on Mondays, Wednesdays, and Fridays, with 3 professors where each professor can teach only one class at a time, and some classes have overlapping time slots.

Time	9:00 - 10:00	9:30 - 10:30	10:00 - 11:00	11:00 - 12:00	11:30 - 12:30
Classes	C1	C2	C3	C4	C5

Professor	Can teach	Other constraint
Ahmed (A)	C2, C4, & C5	
Babar (B)	C1, C2, & C4	
Chishti (C)	C1, C3, & C5	Refuses to teach 2 classes in a row

Draw the Constraint Graph for this problem.

Note: All classes are of exactly one hour duration.